

Operating Systems

Project Report Web Crawler System

Submitted by:

- Ayesha Tariq (BSSE24009)
- Noor Fatima (BSSE24045)
- Mahrukh Khan (BSSE24027)

Submitted to:

- Prof. Umair Shoaib **Date:** May 25, 2026

Performance Metrics

```
===== AGGREGATOR STARTED =====

===== FINAL RESULTS =====
Global Longest Word: transliterated
Global Longest Length: 14

Total Parsed Words: 2851
Average Word Length: 4.78

===== WORD LENGTH DISTRIBUTION =====
Length 1 : 61
Length 2 : 447
Length 3 : 602
Length 4 : 390
Length 5 : 384
Length 6 : 273
Length 7 : 305
Length 8 : 127
Length 9 : 133
Length 10 : 51
Length 11 : 41
Length 12 : 35
Length 13 : 1
Length 14 : 1
Length 15 : 0
Length 16 : 0
Length 17 : 0
Length 18 : 0
Length 19 : 0
Length 20+ : 0

===== PERFORMANCE =====
Downloader Time: 79.6255
Parser Time: 0.0018
Aggregator Time: 0.0002
Total Execution Time: 79.6275
Throughput: 35.80 words/sec

[Aggregator] stats.txt updated successfully.

===== AGGREGATOR FINISHED =====
```

```
===== AGGREGATOR STARTED =====

===== FINAL RESULTS =====
Global Longest Word: transliterated
Global Longest Length: 14

Total Parsed Words: 2851
Average Word Length: 4.78

===== WORD LENGTH DISTRIBUTION =====
Length 1 : 61
Length 2 : 447
Length 3 : 602
Length 4 : 390
Length 5 : 384
Length 6 : 273
Length 7 : 305
Length 8 : 127
Length 9 : 133
Length 10 : 51
Length 11 : 41
Length 12 : 35
Length 13 : 1
Length 14 : 1
Length 15 : 0
Length 16 : 0
Length 17 : 0
Length 18 : 0
Length 19 : 0
Length 20+ : 0

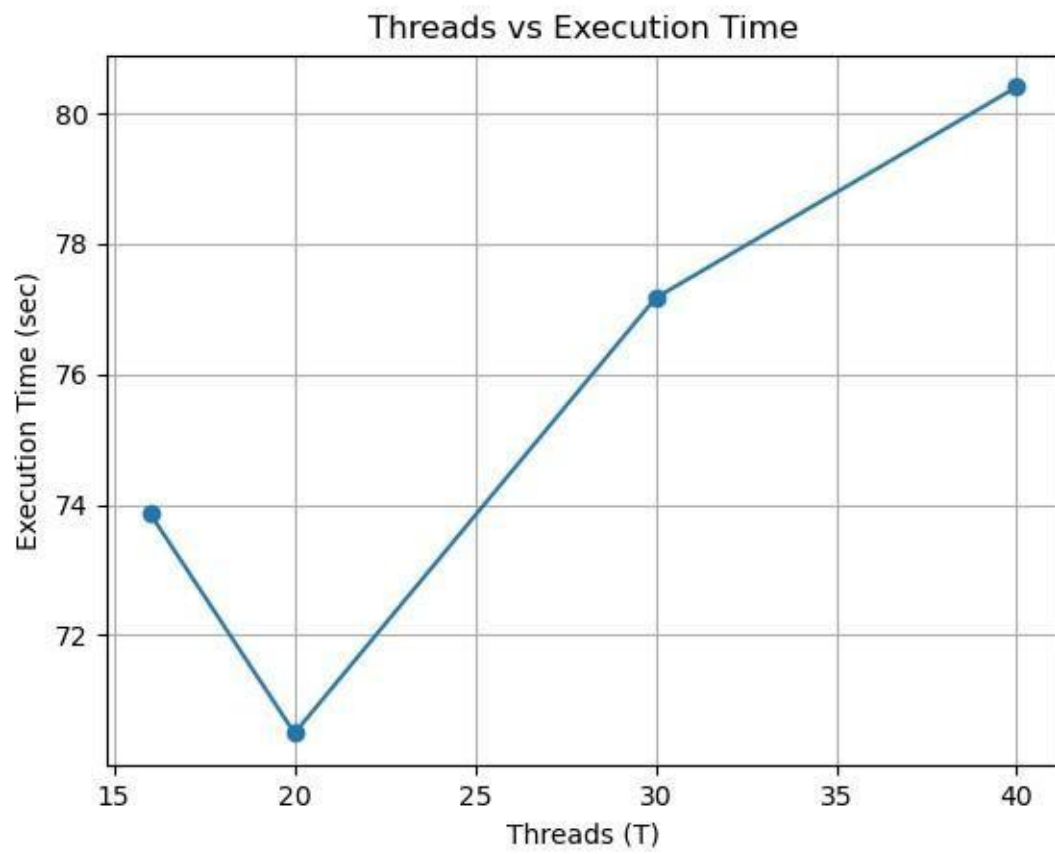
===== PERFORMANCE =====
Downloader Time: 79.6255
Parser Time: 0.0018
Aggregator Time: 0.0002
Total Execution Time: 79.6275
Throughput: 35.80 words/sec

[Aggregator] stats.txt updated successfully.

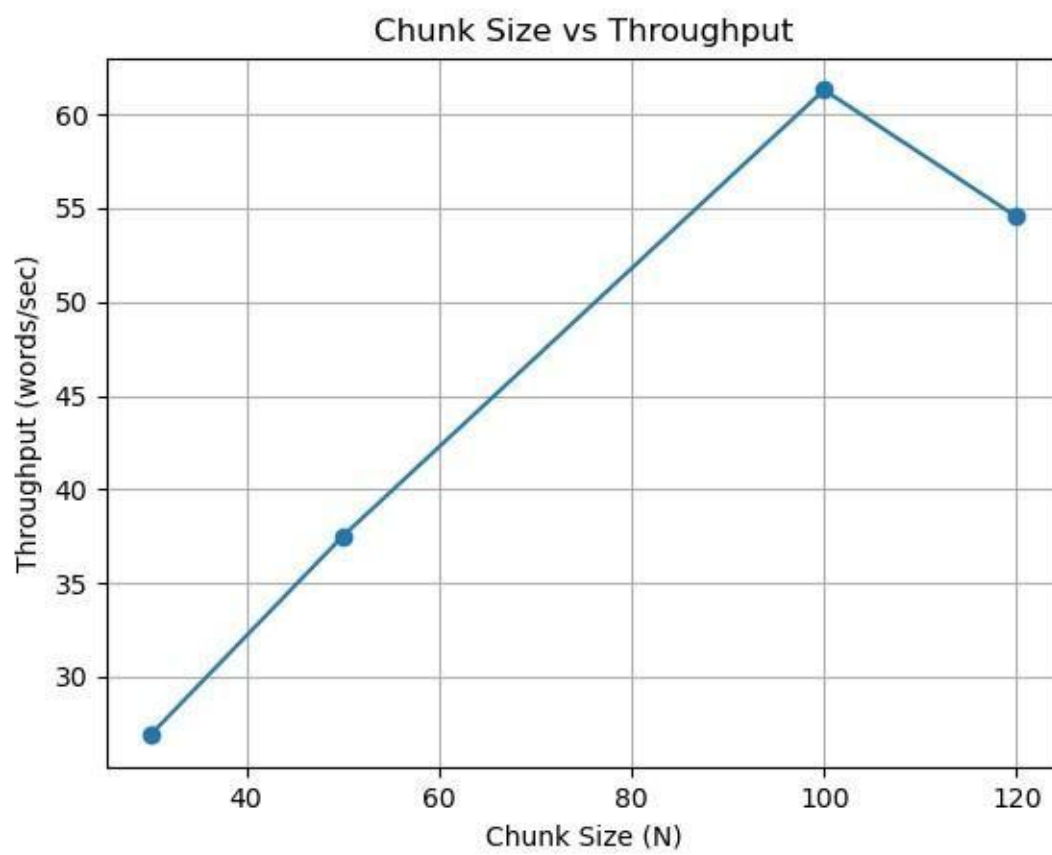
===== AGGREGATOR FINISHED =====
```

Graphs

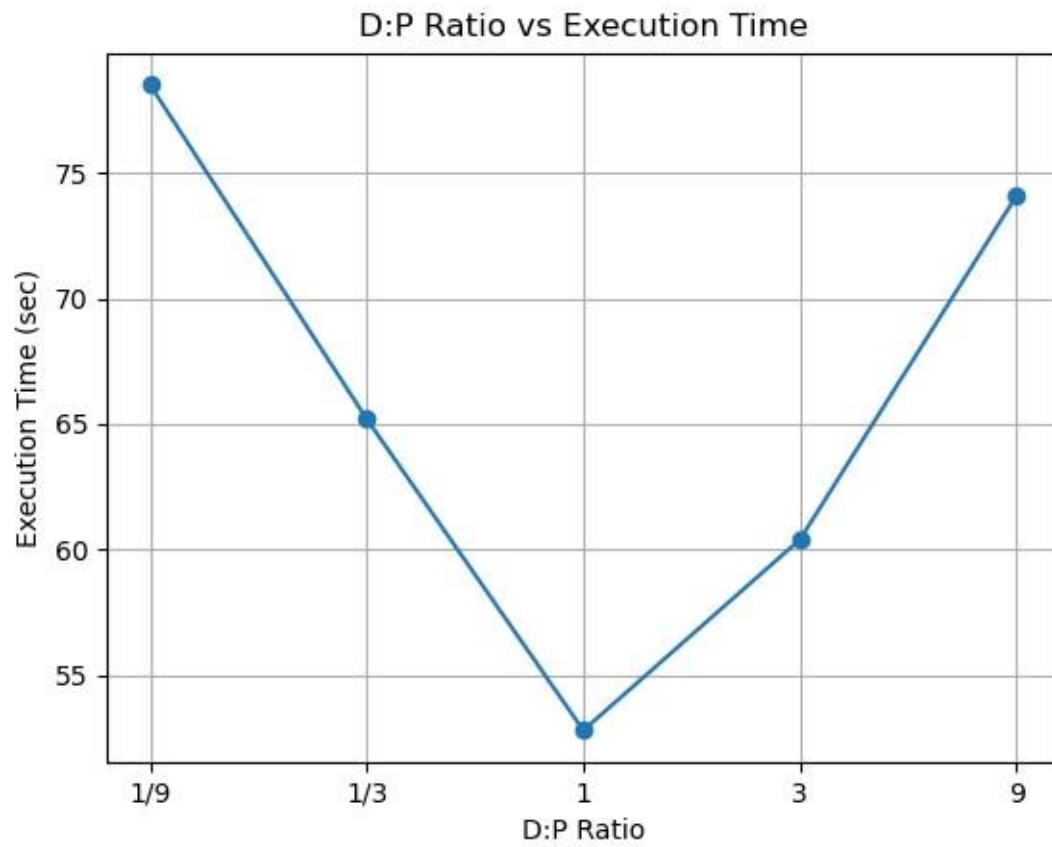
Graph A:



Graph B:



Graph C:



Questions:

a) Which stage is slower: downloading or parsing?

Downloading is significantly slower than parsing.

Why:

- Downloading depends on:
 - Network latency
 - Server response time
 - Bandwidth limitations
- Parsing is CPU-bound and happens in memory, so it is much faster.

In your system:

- Downloader time $\approx$ tens of seconds
- Parser time $\approx$ milliseconds

So the bottleneck is clearly the **downloader stage**.

b) Does increasing threads always improve performance?

No, increasing threads does NOT always improve performance.

Why not:

1. **CPU saturation** ◦ Too many parser threads cause context switching overhead.
2. **Lock contention** ◦ Shared queues and global variables create synchronization delays.
3. **I/O bottleneck (main issue in your system)** ◦ Downloaders depend on network speed, not threads.
4. **Diminishing returns** ◦ After a point, extra threads do not increase throughput.

Conclusion:

Performance improves only up to an optimal thread count
After that, it either stabilizes or degrades

c) Which D:P ratio gives best performance? Why?

Usually best ratio:

$$D = T/2, P = T/2$$

Why balanced ratio works best:

- Downloaders: ○ Keep pipeline filled with data
 - Parsers:
 - Process chunks at similar speed
 - Balanced flow = no starvation
 - No queue buildup
 - Better CPU + I/O overlap
-

Why others are worse:

- **Too many downloaders (D high)**
→ parser becomes bottleneck → queue overflow
 - **Too many parsers (P high)**
→ downloaders cannot supply data fast enough
-

Final insight:

Best performance happens when:

Input rate $\approx$ Processing rate

d) What chunk size gives best performance and why?

Best chunk size is usually:

Medium chunk size (around 25–100 words)

Why:

Small chunk size

- Too many tasks
- High thread overhead

- Too many queue operations

Large chunk size

- Less parallelism
- Load imbalance between threads
- Higher memory usage per chunk

Optimal chunk size

- Good balance of:
 - Parallelism
 - Scheduling overhead
 - CPU utilization

Final conclusion:

Medium chunk size gives best throughput because it balances scheduling overhead and parallel workload distribution.